Worked examples — core methods and further problem types

Table of contents

Worked examples — core methods and further problem types 1
Problem index 2
1. Event algebra and valid probabilities 3
2. Total probability, Bayes and a probability tree 4
3. Multiple causes and event-tree paths 4
4. Normalise a PDF and obtain its CDF and moments 5
5. Piecewise PDF and CDF to PDF 5
6. Binomial counts and at least one 6
7. Poisson counts and independent arrival streams 6
8. Uniform probabilities and quantiles 6
9. Normal tails, quantiles and unknown parameters 7
10. Lognormal moments, tails and survival given age 7
11. Gamma tails, quantiles and expected cost 8
12. Exponential parameter finding and memorylessness 8
13. Weibull lifetime calculations 9
14. Convert reliability to density, hazard and MTTF 9
15. Series systems and minimum lifetime 10
16. Parallel and mixed systems 10
17. Condition on a bridge component 11
18. Complete-data parameter estimation and model choice 11
19. Total time on test and censored exponential estimation 12
20. Repairable systems and interfailure times 12
21. Kaplan–Meier with censoring and tied failures 13
22. Ranks and Weibull probability plots 13
23. Weibull maximum-likelihood estimation 14
24. Normal sample mean, sum and standard error 15
25. Minimum sample size 15
26. Student's t and two-sided critical values 16
27. CLT for Weibull lifetimes 16
28. Shifted Weibull and age-replacement cost 16
29. Engineering calculations and short-answer patterns 17
30. Point mass and continuous lifetime in one model 18
31. A decreasing nonlinear transformation 19
32. Recover lognormal parameters from original-scale moments 19
33. Total variance for a mixture of repair types 20
34. Thinning a Poisson stream 21
35. Competing failure modes and cause-specific probability 21
36. Common-cause failure and the limit of redundancy 22
37. Two-out-of-three with unequal components 22
38. Cold standby and imperfect switching 23
39. Design the number of redundant units 23
40. Survivor selection in a mixed fleet 24
41. Mean remaining life 24
42. Interval censoring and the likelihood 25
43. Delayed entry changes the risk set 25

44. Restricted mean from a survival curve 26
45. Zero failures and a test-exposure requirement 26
46. A distribution-free bound for a sample mean 27
47. Stress-strength interference 28
48. Proof-test spacing and average unavailability 28
49. Inspection timing with a repair lead time 29
50. F–N exceedance frequencies and potential loss of life 29

Cheat sheet · All questions · Engineering notes · R guide

Use the recognition cue to choose a method, then cover the solution and try the calculation. Examples use supplied course questions unless marked **adapted** or **toy**. Extra subparts show closely related skills without repeating whole questions. Numerical results were checked using R; retain full precision until the final answer. In $N(\mu, \sigma^2)$ the second parameter is **variance**.

New practice: Examples 30–50. These are original **toy extensions**, with a different central reasoning step from the worked processes in questions.md and Examples 1–29. Each identifies that new step; familiar probability identities still appear as supporting tools. Some extensions go beyond the supplied midsemester questions, so they are additional practice rather than evidence of examinable scope. The existing course-based examples are retained above them.

Problem index

If the question asks about…	Example
Unions, intersections, disjointness, valid probabilities	1
Overall probability, false positives, reverse conditioning	2
Multiple possible causes / event-tree paths	3
Unknown PDF constant, CDF, mean and variance	4
A piecewise density or a supplied CDF	5
A fixed number of binary trials	6
Counts in a time interval	7
Uniform interval probabilities and quantiles	8
Normal tails, quantiles, parameter finding	9
Lognormal repair time or conditional survival	10
Gamma probabilities / expected fines	11
Exponential rate from a probability, memorylessness	12
Weibull failure probability, hazard, mean, SD, quantile	13
Convert between R, f, h / integrate to find MTTF	14
Failure of any component / minimum lifetime	15
Redundancy / maximum lifetime / system MTTF	16
A bridge network that does not reduce directly	17
Fit complete observations and assess model choice	18
Censored tests / total exposure / units	19
Repairable event timestamps / MTBF / SD	20
Kaplan–Meier steps and censoring	21
Mean ranks and Weibull probability plots	22
Numerically fit Weibull parameters	23

If the question asks about…	Example
Normal sample mean or sum	24
Minimum sample size, one or two tails	25
Student's t and exact versus approximate distributions	26
Non-normal sample average and CLT	27
Scheduled maintenance cost optimisation	28
Discounted costs, availability, engineering judgement	29
A CDF containing a jump and a continuous part	30
Transforming a random speed into a travel time	31
Lognormal parameters from an ordinary mean and SD	32
Variation within and between repair categories	33
Randomly classifying events in a Poisson stream	34
Which failure mode occurs first, and by when	35
Shared failures that defeat independent redundancy	36
Two required survivors with unequal reliabilities	37
A spare that starts ageing only when activated	38
Minimum number of parallel units for a target	39
Falling fleet hazard despite constant individual hazards	40
Expected additional life of an already surviving item	41
Failures only located between inspection times	42
Units entering observation at different ages	43
Mean lifetime limited to an observed horizon	44
A reliability test with no observed failures	45
A probability guarantee without a distribution model	46
Random strength compared with random load	47
Hidden failure averaged over a proof-test interval	48
Detection and repair before a P–F interval ends	49
Constructing an F–N curve and expected annual loss	50

1. Event algebra and valid probabilities

Recognise: "A or B, given C" requires a union inside a conditional probability. **Source:** 2025 paper Q13–14.

Given $P(A) = 0.2$, $P(B) = 0.1$, $P(C) = 0.4$, $A \cap B = \varnothing$, B independent of C, and $P(A \cap C) = 0.1$:

$$P(B \cap C) = P(B)P(C) = 0.04,$$

$$P(A \cup B \mid C) = \frac{P(A \cap C) + P(B \cap C)}{P(C)} = \frac{0.10 + 0.04}{0.40} = 0.35.$$

There is no triple-intersection subtraction because A and B are disjoint. To check any proposed assignment, use

$$\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B)).$$

For $P(A) = 0.5$, $P(B) = 0.7$, the intersection must be at least 0.2: a proposed 0.1 is invalid because it would make the union 1.1. The assignment $0.7, 0.2, 0.1$ is valid and gives union 0.8.

Trap: disjoint events with positive probabilities are not independent. Independence would require a positive intersection.

2. Total probability, Bayes and a probability tree

Recognise: an imperfect test gives conditional probabilities; the observed positive-test frequency can reveal prevalence. **Source:** Prac 1 Q1.

Let A = pipe has corrosion and B = test indicates corrosion. Given $P(B \mid A) = 0.7$, $P(B \mid A') = 0.2$, $P(B) = 0.25$, set $p = P(A)$.

$$0.25 = P(B \mid A)p + P(B \mid A')(1-p) = 0.7p + 0.2(1-p), \qquad p = 0.1.$$

The tree's terminal branches are:

First branch	Second branch	Multiply along the path
Corrosion A: 0.1	Positive B: 0.7	$P(A \cap B) = 0.07$
Corrosion A: 0.1	Negative B': 0.3	$P(A \cap B') = 0.03$
No corrosion A': 0.9	Positive B: 0.2	$P(A' \cap B) = 0.18$
No corrosion A': 0.9	Negative B': 0.8	$P(A' \cap B') = 0.72$

Add the positive branches: $P(B) = 0.07 + 0.18 = 0.25$. Reverse the condition by dividing the appropriate joint probability by the observed event:

$$P(A \mid B) = \frac{0.07}{0.25} = 0.28, \qquad P(A \mid B') = \frac{0.03}{0.75} = 0.04.$$

Trap: $P(B \mid A)$ is not $P(A \mid B)$. The identity to memorise is

$$P(B \mid A)P(A) = P(A \cap B) = P(A \mid B)P(B).$$

The multiplier on the right is $P(B)$.

3. Multiple causes and event-tree paths

Recognise: combine different scenarios, each with its own conditional outcome probability. **Source:** Lecture 1 Q1.

Treat the four stated causes A, B, C, D as mutually exclusive and exhaustive, with probabilities $0.30, 0.40, 0.15, 0.15$. Conditional explosion probabilities are $0.25, 0.20, 0.40, 0.75$.

$$P(E) = \sum_i P(i)P(E \mid i) = 0.075 + 0.080 + 0.060 + 0.1125 = 0.3275.$$

Given that an explosion occurred, the probability assigned to static electricity is

$$P(A \mid E) = \frac{0.075}{0.3275} = 0.2290.$$

Adapted event-tree extension: a leak occurs with probability 0.02 per mission; ignition given a leak has probability 0.3; suppression fails given an ignited leak with probability 0.1. The major-fire path is

$$P(\text{major fire}) = 0.02(0.3)(0.1) = 0.0006 \text{ per mission}.$$

Trap: multiply conditional probabilities along a path and add disjoint outcome paths. Do not sum overlapping possible causes as though they formed a partition. A frequency initiating an event tree produces a frequency at its endpoint; a probability produces a probability.

4. Normalise a PDF and obtain its CDF and moments

Recognise: the density contains an unknown constant. **Adapted from:** Stats Quiz 1 Q8, replacing its known coefficient by c.

Let $f(x) = \mathrm{cx}$ for $0 < x < 8$, zero otherwise. Total area must be one:

$$1 = \int_0^8 \mathrm{cx}\,\mathrm{dx} = 32c \quad \Rightarrow \quad c = \frac{1}{32}.$$

Then

$$F(x) = \begin{cases} 0 & x \le 0 \\ \frac{x^2}{64} & 0 < x < 8 \\ 1 & x \ge 8. \end{cases}$$

$$P(1.6 < X < 4.1) = \frac{4.1^2 - 1.6^2}{64} = 0.22265625 \approx 0.22.$$

$$E[X] = \int_0^8 \frac{x^2}{32}\,\mathrm{dx} = \frac{16}{3}, \qquad E[X^2] = \int_0^8 \frac{x^3}{32}\,\mathrm{dx} = 32,$$

$$\mathrm{Var}(X) = 32 - \left(\frac{16}{3}\right)^2 = \frac{32}{9}, \qquad \mathrm{SD}(X) = \frac{\sqrt{32}}{3} \approx 1.8856.$$

For the median, $F(m) = 0.5$ gives $m = \sqrt{32} \approx 5.6569$.

Trap: a density height is not a point probability; $P(X = x) = 0$ for a continuous variable. Variance requires subtracting the squared mean.

5. Piecewise PDF and CDF to PDF

Recognise: the integration rule changes at a breakpoint. **Sources:** Lecture 1 Q2, Stats Quiz 1 Q7.

Repair time has $f(x) = \frac{x}{1600}$ for $0 < x \le 40$ and $\frac{80-x}{1600}$ for $40 < x < 80$, zero elsewhere. Integrating from the beginning of the support gives

$$F(x) = \begin{cases} 0 & x \le 0 \\ \frac{x^2}{3200} & 0 < x \le 40 \\ 1 - \frac{(80-x)^2}{3200} & 40 < x < 80 \\ 1 & x \ge 80. \end{cases}$$

The second branch includes the area already accumulated to 40. Therefore

$$P(60 \le X \le 70) = F(70) - F(60) = \frac{20^2 - 10^2}{3200} = 0.09375.$$

For an interval crossing the breakpoint (added subpart),

$$P(30 < X < 50) = F(50) - F(30) = 1 - \frac{30^2}{3200} - \frac{30^2}{3200} = 0.4375.$$

Symmetry gives $E[X] = 40$ h; equivalently integrate $xf(x)$ separately over $(0, 40)$ and $(40, 80)$.

Reverse direction: if $F(x) = \frac{(x+1)^2}{9}$ on $-1 < x \le 2$, with zero below and one above, then $f(x) = F'(x) = 2\frac{x+1}{9}$ on $(-1, 2)$ and zero elsewhere. Check the CDF is continuous at the boundaries; a jump would represent point mass.

Trap: do not start the second CDF branch at zero. Cumulative probability does not reset at a breakpoint.

6. Binomial counts and at least one

Recognise: fixed n, independent trials, two outcomes and constant p. **Source:** Prac 2 Q1.

For ten resistors with defect probability 0.08, $X \sim \text{Binomial}(10, 0.08)$:

$$P(X > 1) = 1 - P(X = 0) - P(X = 1) = 1 - 0.92^{10} - 10(0.08)(0.92)^9 = 0.1879.$$

Added subparts: $E[X] = \text{np} = 0.8$, $\text{SD}(X) = \sqrt{np(1-p)} = \sqrt{0.736} \approx 0.8579$. The probability of at least one defect is $1 - 0.92^{10} \approx 0.5656$.

```
pbinom(1, size = 10, prob = 0.08, lower.tail = FALSE)
```

Trap: $P(X \ge k) = 1 - P(X \le k - 1)$. Discrete endpoints matter.

7. Poisson counts and independent arrival streams

Recognise: events counted over time at a specified rate. **Sources:** 2025 Q15, Prac 2 Q2.

Repair requests arrive at one per week. Over four weeks $N \sim \text{Poisson}(4)$, so

$$P(N = 3) = \frac{4^3 e^{-4}}{3!} = 0.195367 \approx 0.195.$$

The parameter is rate times interval, $\mu = \lambda t$, and both mean and variance equal 4.

For independent elevator/passenger streams at rates 1 and 5 per minute, respectively, the chance of no elevator and at least ten passengers in one minute is

$$e^{-1}\left[1 - \sum_{k=0}^{9} \frac{5^k e^{-5}}{k!}\right] = 0.011709.$$

```
dpois(0, 1) * ppois(9, 5, lower.tail = FALSE)
```

Trap: the rate is not automatically the Poisson parameter. Independence of the two arrival streams is an assumption needed for the product.

8. Uniform probabilities and quantiles

Recognise: constant density over a finite interval. **Source:** Lecture 2, hose model $X \sim U(85, 165)$.

The PDF height is $\frac{1}{165-85} = \frac{1}{80}$. Thus

$$P(X < 120) = \frac{120 - 85}{80} = 0.4375.$$

For 99% survival to age c,

$$P(X > c) = \frac{165 - c}{80} = 0.99 \quad \Rightarrow \quad c = 85.8 \text{ h}.$$

Added summaries: $E[X] = 125$ h, $\mathrm{SD}(X) = \frac{80}{\sqrt{12}} \approx 23.0940$ h. On $85 < t < 165$, $R(t) = \frac{165-t}{80}$ and $h(t) = \frac{1}{165-t}$: a flat density does not mean constant hazard.

Trap: clip probability intervals to the support; the CDF is zero below 85 and one above 165.

9. Normal tails, quantiles and unknown parameters

Recognise: standardise a boundary for a probability; invert a CDF for a boundary. **Sources:** Lecture 2, Stats Quiz 2 Q2.

For $X \sim N(125, 14^2)$,

$$P(X < 120) = \Phi\left(\frac{120 - 125}{14}\right) = 0.3605.$$

For $P(X > c) = 0.99$, only 0.01 lies below c:

$$c = 125 + 14\Phi^{-1}(0.01) = 92.43 \text{ h}.$$

For $P(|Z| > z) = 0.85$, each tail is 0.425, so $z = \Phi^{-1}(0.575) = 0.1891 \approx 0.19$.

Adapted parameter-finding subpart: if $X \sim N(\mu, 10^2)$ and $P(X < 400) = 0.05$, then $\frac{400-\mu}{10} = z_{0.05} = -1.64485$, giving $\mu = 416.4485$.

More generally two known quantiles x_1, x_2 at probabilities p_1, p_2 give

$$\sigma = \frac{x_2 - x_1}{z_{p_2} - z_{p_1}}, \qquad \mu = x_1 - \sigma z_{p_1}.$$

Trap: R's `sd` is the standard deviation. A normal lifetime model allows negative values; check whether that is negligible for the application.

10. Lognormal moments, tails and survival given age

Recognise: $\ln T$ is normal or $T = \exp(X)$ for normal X. **Sources:** 2025 Q16, Prac 3 Q5.

For a repair time $T = e^X$ with $X \sim N(1.2, 4.3)$,

$$P(T > 4) = P(X > \ln 4) = 1 - \Phi\left(\frac{\ln 4 - 1.2}{\sqrt{4.3}}\right) = 0.464207 \approx 0.464.$$

For the light-bulb problem $\ln T \sim N(10, 0.2^2)$ (log-scale parameters; T measured in hours):

$$E[T] = e^{10 + \frac{0.2^2}{2}} = 22471.43 \text{ h},$$

$$\mathrm{SD}(T) = e^{10 + \frac{0.2^2}{2}} \sqrt{e^{0.2^2} - 1} = 4539.61 \text{ h}.$$

Also $E[\ln T] = 10$ and $\mathrm{Var}(\ln T) = 0.04$. The probability of surviving to 30000 h is

$$R(30000) = 1 - \Phi\left(\frac{\ln 30000 - 10}{0.2}\right) = 0.061202.$$

Given survival to 20000 h, survival to a **total age** of 30000 h has probability

$$\frac{R(30000)}{R(20000)} = 0.089307.$$

Trap: "another 30000 h after 20000 h" would instead require $\frac{R(50000)}{R(20000)}$. The lognormal is not memoryless; e^{μ} is its median, not its mean.

11. Gamma tails, quantiles and expected cost

Recognise: a positive gamma variable, often evaluated by software. **Sources:** Stats Quiz 2 Q5–6, Prac 1 Q3.

With shape $\alpha = 2.27$ and rate $\lambda = 1.22$,

```
pgamma(2.48, shape = 2.27, rate = 1.22, lower.tail = FALSE)
# 0.2506520
```

Hence $P(X > 2.48) = 0.251$. For shape 2.26, rate 1.27, and $P(X \le c) = 0.78$:

```
qgamma(0.78, shape = 2.26, rate = 1.27)
# 2.519904
```

For the effluent model $X \sim \Gamma(2, \text{rate} = 1)$, integration by parts gives

$$f(x) = x\mathrm{e}^{-x}, \qquad F(x) = \int_0^x u\mathrm{e}^{-u}\,\mathrm{d}u = 1 - (1+x)e^{-x}, \quad x > 0.$$

Mean and variance are both 2. A test exceeds threshold a with probability $(1+a)e^{-a}$. If each exceedance costs C, two weekly tests at threshold 4 have expected fortnightly cost

$$2C(1+4)e^{-4} = 0.183156C,$$

whereas one fortnightly test at threshold 3 costs $4C\mathrm{e}^{-3} = 0.199148C$ on average. The first has lower expected cost under these assumptions.

Trap: `scale = 1/rate`. The expected number of fines can be fractional; linearity of expectation does not require independent tests.

12. Exponential parameter finding and memorylessness

Recognise: constant hazard or an exponential lifetime, with a survival probability supplied. **Source:** Week 5 Quiz Q3 added summaries and conditional-survival subparts.

Given $R(1000) = 0.46$,

$$0.46 = e^{-1000\lambda} \quad \Rightarrow \quad \lambda = -\frac{\ln 0.46}{1000} = 0.000776529 \text{ h}^{-1}.$$

$$P(T \le 336) = 1 - e^{-336\lambda} = 0.229653 \approx 0.230.$$

Mean life and standard deviation are both $\frac{1}{\lambda} = 1287.78$ h; median life is $\ln \frac{2}{\lambda} = 892.62$ h.

For an item already aged 1000 h,

$$P(T > 1336 \mid T > 1000) = \frac{e^{-1336\lambda}}{e^{-1000\lambda}} = e^{-336\lambda} = 0.770347.$$

Its chance of failing in the next 336 h is again 0.229653. By contrast, the unconditional chance of failing between ages 1000 and 1336 is $R(1000) - R(1336) = 0.46(0.229653) \approx 0.105640$.

Trap: an unconditional interval probability is a difference of survivals; a conditional probability divides by survival to the starting age. “Good as new after repair” alone does not establish exponential lifetimes.

13. Weibull lifetime calculations

Recognise: given shape β and characteristic life η. **Sources:** Prac 3 Q3, Lecture 3, 2025 Q8.

For solar cells with $\beta = 2.5$, $\eta = 10$ years:

$$R(t) = e^{-\left(\frac{t}{10}\right)^{\{2.5\}}}, \qquad h(t) = \frac{2.5}{10}\left(\frac{t}{10}\right)^{1.5}.$$

Age	Hazard, year^{-1}
5 yr	0.088388
10 yr	0.250000
20 yr	0.707107

The increasing hazard represents wear-out. The mean and standard deviation are

$$E[T] = 10\Gamma(1.4) = 8.87264 \text{ yr},$$

$$\mathrm{SD}(T) = 10\sqrt{\Gamma(1.8) - \Gamma(1.4)^2} = 3.79667 \text{ yr}.$$

The supplied rounded gamma values yield slightly different final decimals. Use the precision requested by the question.

For the 2025 foundation model, $\beta = 1.2$, $\eta = 14$ yr:

$$F(39) = 1 - e^{-\left(\frac{39}{14}\right)^{\{1.2\}}} = 0.967261 \approx 0.967.$$

For any Weibull, the pth failure quantile is $t_p = \eta[-\ln(1-p)]^{\frac{1}{\beta}}$. The lecture’s $\beta = 0.5, \eta = 125$ model gives median $125(\ln 2)^2 = 60.06$ h, mean $125\Gamma(3) = 250$ h, and decreasing hazard. In both models $F(\eta) = 1 - e^{-1} = 0.632121$.

Adapted residual-life check: for $\beta = 2, \eta = 5$, compare

$$P(T > 6 \mid T > 4) = e^{-\left[\left(\frac{6}{5}\right)^2 - \left(\frac{4}{5}\right)^2\right]} = e^{-0.8} = 0.449329$$

with $P(T > 2) = e^{-0.16} = 0.852144$. The Weibull is memoryless only at $\beta = 1$.

Trap: η is neither generally the mean nor the median. Hazard is a rate conditional on survival, not a failure probability.

14. Convert reliability to density, hazard and MTTF

Recognise: a function is supplied instead of a named distribution. **Source:** Stats Quiz 3 Q1, Q2, Q10.

With $R(t) = \frac{3.24}{[(t+2.7)(t+1.2)]}$, $t \geq 0$, differentiate:

$$f(t) = -R'(t) = \frac{3.24(2t + 3.9)}{(t + 2.7)^2(t + 1.2)^2}, \qquad f(2.2) = 0.096889 \approx 0.10.$$

For $R(t) = \frac{2.08}{[(t+1.6)(t+1.3)]}$, logarithmic differentiation is easier:

$$h(t) = -\frac{d}{\mathrm{dt}} \ln R(t) = \frac{1}{t + 1.6} + \frac{1}{t + 1.3}, \qquad h(2) = 0.580808 \approx 0.58.$$

For $R(t) = (1+t)^{-7}$,

$$\text{MTTF} = \int_0^\infty R(t)\,\mathrm{dt} = \left[-\frac{(1+t)^{-6}}{6}\right]_0^\infty = \frac{1}{6} \approx 0.17.$$

Adapted reverse-direction check: if $h(t) = 2\frac{t}{9}$, then $H(t) = \int_0^t 2\frac{u}{9}\,\mathrm{du} = \frac{t^2}{9}$, $R(t) = e^{-\frac{t^2}{9}}$, $F(t) = 1 - R(t)$ and $f(t) = \left(2\frac{t}{9}\right)e^{-\frac{t^2}{9}}$. This is Weibull ($\beta = 2, \eta = 3$).

Trap: $f = \text{hR}$, not $F = \text{hR}$. Check $R(0) = 1$, non-increasing survival, and total density area one. The survival-integral MTTF formula assumes nonnegative lifetimes.

15. Series systems and minimum lifetime

Recognise: the system fails as soon as any required component fails. **Sources:** Week 5 Quiz Q4, Prac 2 Q7, Week 4 quiz.

For independent exponential component lifetimes with mean lives $15020, 1200, 10000, 11000$ h,

$$\lambda_{\text{sys}} = \sum_i \lambda_i = \frac{1}{15020} + \frac{1}{1200} + \frac{1}{10000} + \frac{1}{11000},$$

$$R_{\text{sys}}(500) = e^{-500\lambda_{\{\text{sys}\}}} = 0.579604 \approx 0.580.$$

The same argument handles $Y = \min(X_1, X_2, X_3)$ with rates $0.002, 0.003, 0.005$: $Y \sim \text{Exp}(0.010)$, $P(Y > 300) = e^{-3} = 0.049787$, and $E[Y] = 100$ h. The Prac 2 solution's printed 0.0005 in one product is inconsistent with its stated 0.005 and numerical result.

Units/checking variant: five sensors each at 1.91 failures per million hours give $\lambda_{\text{module}} = 9.55 \times 10^{-6}$ h^{-1}. Using 365.25 days/year,

$$R(10\text{ yr}) = e^{-9.55\times10^{\{-6\}}(10)(365.25)(24)} = 0.432941.$$

The local Week 4 quiz lists 0.426 as its selected answer, but that does not follow exactly from its stated 1.91 rate. The workshop data instead give about 1.95 per million h; using 1.95 and 365 days/year gives 0.425666, explaining the selected option. See the engineering notes for the source-data calculation. Its stated whole-system rate 15.44 per million h gives mean life $\frac{10^6}{15.44} = 64766.84$ h $= 7.3884$ yr, below the ten-year requirement.

Trap: multiply reliabilities, not component mean lives. Sum hazards for independent series components; summing constant rates yields an exponential system only when the component hazards are constant.

16. Parallel and mixed systems

Recognise: redundancy means the subsystem fails only if all alternative paths fail. **Sources:** 2025 Q4, Stats Quiz 4 Q1.

Three independent a components in parallel, then b and c in series, with $a = 0.62, b = 0.91, c = 0.70$:

$$R_a = 1 - (1-a)^3 = 1 - 0.38^3,$$

$$R_{\text{sys}} = R_a\, b\, c = (1 - 0.38^3)(0.91)(0.70) = 0.602047 \approx 0.60.$$

For independent exponential A, B in parallel and C in series, rates $a = 0.14, b = 0.04, c = 0.23$:

$$R_{\text{sys}}(t) = \left[e^{-\text{at}} + e^{-\text{bt}} - e^{-(a+b)t}\right]e^{-\text{ct}}.$$

The parallel lifetime is $\max(T_A, T_B)$; the complete system lifetime is $\min(\max(T_A, T_B), T_C)$. Integrating survival gives

$$E\big[T_{\text{sys}}\big] = \frac{1}{a+c} + \frac{1}{b+c} - \frac{1}{a+b+c} = 3.967382.$$

Trap: the final exponential term is subtracted. Adding all three gives $R(0) = 3$, impossible. A parallel combination of exponentials is generally not exponential, so $\frac{1}{h(t)}$ is not generally its MTTF.

17. Condition on a bridge component

Recognise: a cross-link prevents immediate series/parallel reduction. **Toy example for the conditional-network method in the engineering reliability material.**

Five independent links have reliability 0.9. Two paths connect source S to destination D: upper $S \to U \to D$ has links A, B; lower $S \to V \to D$ has C, D_1. A bidirectional bridge link E joins U and V.

When E fails, the two series paths act in parallel:

$$R_{\text{sys}\mid E'} = 1 - (1 - R_A R_B)\big(1 - R_C R_{D_1}\big) = 1 - \left(1 - 0.9^2\right)^2 = 0.9639.$$

When E works, U and V effectively merge. The input links A, C are parallel, followed in series by parallel output links B, D_1:

$$R_{\text{sys}\mid E} = \left[1 - (1 - 0.9)^2\right]^2 = 0.9801.$$

Total probability gives

$$R_{\text{sys}} = R_E R_{\text{sys}\mid E} + (1 - R_E) R_{\text{sys}\mid E'} = 0.9(0.9801) + 0.1(0.9639) = 0.97848.$$

Trap: paths sharing a component are not independent, even if components are independent. Condition on the shared component or bridge, simplify each state, then combine.

18. Complete-data parameter estimation and model choice

Recognise: observed times are data, not supplied population parameters. **Source:** Lecture 4, fabricated hose repair data in the Westralia teaching case.

For $x = (10, 17, 24, 30, 14, 18, 15, 13, 21)$ minutes, $n = 9$ and $\bar{x} = 18$.

Model	Maximum-likelihood estimates
Normal	$\hat{\mu} = 18$; $\hat{\sigma} = \sqrt{\frac{\sum (x_i - 18)^2}{9}} = 5.811865$
Lognormal	$\hat{\mu} = \text{mean}(\ln x) = 2.840222$; $\hat{\sigma} = \sqrt{\frac{\sum (\ln x_i - \hat{\mu})^2}{9}} = 0.315505$
Exponential	$\hat{\lambda} = \frac{n}{\sum} x_i = \frac{1}{18} = 0.055556 \text{ min}^{-1}$

The ordinary sample standard deviation is $s = \sqrt{\frac{\sum (x_i - 18)^2}{8}} = 6.164414$ min. Report which estimator is requested; R's `sd(x)` uses $n - 1$.

For model checking, plot a density histogram and fitted density; check positivity, skewness and tail fit. Here $\frac{s}{\bar{x}} = 0.3425$ is far below the exponential model's population SD/mean of 1. That is a reason to question exponential fit, although nine observations cannot decisively identify the true distribution.

Simulation and graphical-summary subpart: Stats Quiz 1 Q5 asks for a reproducible sample of twenty $N(10, 5^2)$ observations. Generate once and retain the vector:

```r
set.seed(5507)
x <- rnorm(20, mean = 10, sd = 5)
c(mean = mean(x), median = median(x), sd = sd(x))
# mean 11.366560; median 11.349722; sd 6.036354
hist(x, probability = TRUE)
curve(dnorm(x, mean = 10, sd = 5), add = TRUE, col = "red")
boxplot(x)
```

The requested median is 11.350. The realised sample mean and SD differ from the population values 10 and 5 because this is one finite random sample. A density histogram, rather than raw counts, makes the theoretical density overlay comparable. Resetting the seed reproduces the sample; repeatedly calling `rnorm` without resetting generates new data.

Trap: lognormal parameters are estimated from **logs of the observations**, not the log of the sample mean. A histogram suggests candidate models; it does not prove the hazard shape.

19. Total time on test and censored exponential estimation

Recognise: surviving units contribute time but no failure. **Source:** 2025 Q3 and Q10.

Seven systems start together. Four fail at $3, 13, 25, 40$ months; three survive to 48 months without replacement. Total exposure is

$$\mathcal{T} = 3 + 13 + 25 + 40 + 3(48) = 225 \text{ component-months.}$$

With four failures and an exponential model,

$$\hat{\lambda} = \frac{4}{\left(\frac{225}{12}\right)(365.25)(24)} = 24.33645 \text{ failures per million h.}$$

The fitted mean life is $\frac{\mathcal{T}}{4} = 56.25$ months. This is a model-based estimate incorporating censored exposure, not the average of the four observed failure ages.

Failed-test variant: 113 foundation tests include five unsuccessful equipment tests with no usable durations supplied. Excluding these leaves 108 valid foundations: five fail at $3, 5, 6, 8, 12$ months and 103 survive to 24 months.

$$\mathcal{T} = 3 + 5 + 6 + 8 + 12 + 103(24) = 2506 \text{ months.}$$

The answer follows the paper's exclusion convention. If usable observation durations for the unsuccessful tests were known, they would require a stated censoring/data-quality decision.

Trap: do not use 7×48 when failed units leave the test, and do not divide failures by calendar duration alone when multiple units are observed. $\sum \frac{t_i}{r}$ estimates MTTF under an exponential model; it is not a universal censored-data mean estimator.

20. Repairable systems and interfailure times

Recognise: timestamps describe repeated failures of one operating asset. **Source:** Stats Quiz 4 Q6–7.

Five failures occur at operating times $1, 2, 3, 5, 8$ and observation continues to 17.5. Total operating exposure is 17.5, so

$$\widehat{\text{MTBF}} = \frac{17.5}{5} = 3.5.$$

For the separate timestamps $2, 3, 6, 9, 10$, observed to 11, completed interfailure durations are

$$
\begin{aligned}
&2,\\
&3-2,\\
&6-3,\\
&9-6,\\
&10-9=2,1,3,3,1.
\end{aligned}
$$

Their mean is 2. The population-divisor descriptive SD used in the local answer is $\sqrt{\frac{4}{5}}=0.894427$. The conventional sample SD, which R's `sd` returns, is $\sqrt{\frac{4}{4}}=1$.

The final one-unit interval is right-censored, so it is not an observed complete duration. Under an exponential renewal model, all eleven exposure units give fitted mean $\frac{11}{5}=2.2$ and fitted lifetime SD 2.2, a different quantity from the descriptive SD of completed intervals.

Trap: do not take the SD of absolute timestamps. Include final censored exposure in an exposure-based rate estimate. If times are calendar time with repair downtime, establish operating time before using it as exposure.

21. Kaplan–Meier with censoring and tied failures

Recognise: estimate survival without selecting a parametric lifetime distribution. **Toy dataset illustrating** Prac 4 Q3 the LMS `exa1.csv` is not supplied locally.

Five units have observations $(2,F),(3,C),(4,F),(4,F),(5,C)$, where F means failure and C censoring.

Time	At risk just before time	Failures	Censored	Survival after time
2	5	1	0	$1\left(1-\frac{1}{5}\right)=0.8$
3	4	0	1	0.8
4	3	2	0	$0.8\left(1-\frac{2}{3}\right)=0.266667$
5	1	0	1	0.266667

Thus $\hat{R}(3.5)=0.8$, $\hat{R}(4)=0.266667$ and $\hat{F}(4)=0.733333$. Censoring changes the next risk set but does not itself cause a drop.

```
library(survival)
d <- data.frame(time = c(2,3,4,4,5), status = c(1,0,1,1,0))
fit <- survfit(Surv(time, status) ~ 1, data = d)
summary(fit, times = c(3.5, 4))
```

Trap: using $1-\frac{3}{5}=0.4$ ignores censoring before the final failures. A positive KM tail beyond the last observation does not identify the full mean lifetime. Standard KM relies on non-informative censoring; recorded suspensions are not automatically harmless.

22. Ranks and Weibull probability plots

Recognise: the question explicitly asks for a plotting position or parameters from a straight line.
Sources: Stats Quiz 4 Q3, Prac 3 Q4.

Eleven units are tested until the ninth failure at 13.48. The requested **mean rank** is

$$
\hat{F}(t_9)=\frac{9}{11+1}=0.75.
$$

For comparison, the empirical/KM fraction with no earlier censoring would be $\frac{9}{11} = 0.818182$, while the approximate median rank is $\frac{9-0.3}{11+0.4} = 0.763158$. These are different estimators, so use the named method.

For Weibull data, define $x = \ln t$ and $y = \ln[-\ln(1 - F(t))]$. Then

$$y = \beta x - \beta \ln \eta.$$

Toy fitted line: $y = 2x - 3.218876$. Reading slope and intercept gives $\hat{\beta} = 2$ and

$$\hat{\eta} = e^{-\frac{-3.218876}{2}} \approx 5.$$

Straightness supports a two-parameter Weibull fit; curvature can indicate a shift, mixture or poor model choice. In the supplied mixed plot in Week 5 Quiz Q6, the earlier segment has $\beta_1 = 1.5278 > \beta_2 = 1.3554$.

Adjusted median-rank subpart with suspensions: a toy dataset following the method in the Weibull slides is $(2, F), (3, C), (4, F), (5, F), (6, C)$. Here $n = 5$. Start with adjusted rank $j_{\text{prev}} = 0$; for each failure let m count all preceding observations and calculate

$$\text{INC} = \frac{n + 1 - j_{\text{prev}}}{1 + n - m}, \qquad j = j_{\text{prev}} + \text{INC}, \qquad \hat{F} = \frac{j - 0.3}{n + 0.4}.$$

Failure time	m	Increment	Adjusted rank j	Median-rank estimate
2	0	$\frac{6-0}{6} = 1$	1	$\frac{0.7}{5.4} = 0.129630$
4	2	$\frac{6-1}{4} = 1.25$	2.25	$\frac{1.95}{5.4} = 0.361111$
5	3	$\frac{6-2.25}{3} = 1.25$	3.50	$\frac{3.2}{5.4} = 0.592593$

The suspension at time 3 receives no failure point but increases m for the next update. Continue from the previous **adjusted** rank rather than resetting to the observed failure number. These are plotting positions, distinct from Kaplan–Meier estimates.

Trap: check which variable is on each axis; swapping axes changes the slope. Do not transform $F = 0$ or $F = 1$ because the logs diverge. With interspersed censoring use a censoring-aware CDF estimator, not unadjusted failure ranks.

23. Weibull maximum-likelihood estimation

Recognise: asked to fit shape and scale numerically. **Source:** Stats Quiz 4 Q10, which explicitly instructs omission of censored values.

Use these twenty observed failures:

```
x <- c(8,21.3,25.5,8.1,4.5,4.2,13.2,9.9,27,25.7,
       25.8,28.6,13.5,7,20.9,9.6,33.9,16.5,4.4,19.2)
score <- function(beta) {
  1/beta + mean(log(x)) - sum(x^beta * log(x))/sum(x^beta)
}
beta_hat <- uniroot(score, c(0.1, 10), tol = 1e-12)$root
eta_hat <- mean(x^beta_hat)^(1/beta_hat)
c(shape = beta_hat, scale = eta_hat)
# shape 1.888809; scale 18.459323
```

The score equation is

$$\frac{1}{\beta}+\frac{1}{n}\sum \ln x_i-\frac{\sum x_i^{\beta}\ln x_i}{\sum x_i^{\beta}}=0,$$

with $\hat{\eta}=\left[\sum \frac{x_i^{\hat{\beta}}}{n}\right]^{\frac{1}{\hat{\beta}}}$. Therefore the requested shape is 1.889.

If censored observations are to be retained, use their survival contribution. For observed times t_i and failure indicators δ_i:

$$\ell(\beta,\eta)=\sum_i \delta_i[\ln\beta-\beta\ln\eta+(\beta-1)\ln t_i]-\sum_{i\left(\frac{t_i}{\eta}\right)}^{\beta}.$$

Trap: omitting suspensions is required for this particular quiz calculation, but is generally a biased way to fit censored lifetime data. Never relabel a suspension as a failure. See the R guide for the censoring-aware fit.

24. Normal sample mean, sum and standard error

Recognise: the random variable is an average of n measurements. **Source:** 2025 Q17 sum subpart added from the Prac 5 method.

Loaded bucket volumes are iid $N(11.5,1.4^2)$ m^3. For $n=12$:

$$\bar{X}\sim N\left(11.5,\frac{1.4^2}{12}\right),\qquad \mathrm{SE}(\bar{X})=\frac{1.4}{\sqrt{12}}.$$

$$P(\bar{X}<10.88)=\Phi\left(\frac{10.88-11.5}{\frac{1.4}{\sqrt{12}}}\right)=0.062502\approx 0.063.$$

For total volume $S=\sum X_i$, $S\sim N(138,23.52)$, and $P(S<130.56)$ is the same probability because $130.56=12(10.88)$.

Trap: the sample mean uses SD $\frac{\sigma}{\sqrt{n}}$; the sum uses SD $\sigma\sqrt{n}$. Both results are exact for independent normal observations, even when $n<30$.

25. Minimum sample size

Recognise: find n to meet a probability or precision requirement. **Sources:** Stats Quiz 5 Q4, Lecture 5.

Normal observations have variance 16, so $\sigma=4$. Require $P(|\bar{X}-\mu|<2)\geq 0.99$. The middle 99% leaves 0.005 in each tail:

$$\frac{2}{\frac{4}{\sqrt{n}}}\geq z_{0.995}=2.575829,\qquad n\geq\left(\frac{4z_{0.995}}{2}\right)^2=26.539586.$$

Hence $n_{\min}=27$. Check: $n=26$ gives 0.989213, below the requirement; $n=27$ gives 0.990625, above it.

For the lecture's **one-sided** requirement with mean 300, SD 15, and $P(\bar{T}\geq 310)\leq 0.005$:

$$n\geq\left(\frac{15z_{0.995}}{310-300}\right)^2=14.928517,\qquad n_{\min}=15.$$

Trap: the first problem divides its total tail probability by two; the second already specifies one tail. Always round the required sample size up.

26. Student's t and two-sided critical values

Recognise: unknown population SD is estimated by S, or a t distribution is explicitly given. **Sources:** 2025 Q18, Stats Quiz 5 Q5.

For twenty independent normal lifetimes with unknown mean and variance,

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{20}}} \sim t_{19} \text{ exactly.}$$

For $T \sim t_5$, find positive c satisfying $P(|T| > c) = 0.0125$. Each tail has area 0.00625, so

$$c = t_{5,0.99375} = 3.810005 \approx 3.810.$$

```
qt(1 - 0.0125/2, df = 5)
2 * pt(3.810004700, df = 5, lower.tail = FALSE) # about 0.0125
```

Trap: normality of the population makes the t result exact. A large sample from an unspecified distribution supports approximate normal inference, not an exact t law. When σ is known, the normal-population standardised mean uses Z, not t.

27. CLT for Weibull lifetimes

Recognise: the individual distribution is non-normal, but the question concerns a mean of many independent observations. **Source:** Prac 5 Q5.

For $T_i \sim \text{Weibull}(2, 5)$ and $n = 50$, first calculate the individual moments:

$$\mu = 5\Gamma(1.5) = 4.431135, \quad \sigma^2 = 25\left[\Gamma(2) - \Gamma(1.5)^2\right] = 5.365046.$$

For **one** board, $P(T_1 > 5) = e^{-1} = 0.367879$. For the **sample average**, the CLT gives

$$\bar{T} \dot{\sim} N\left(4.431135, \frac{5.365046}{50}\right),$$

$$P(\bar{T} > 5) \approx 1 - \Phi\left(\frac{5 - 4.431135}{\sqrt{\frac{5.365046}{50}}}\right) = 0.041226.$$

For $P(|\bar{T} - \mu| < 0.5) > 0.95$, the CLT sample-size calculation is

$$n > \left(\frac{z_{0.975}\sqrt{5.365046}}{0.5}\right)^2 = 82.438412, \qquad n_{\min} \approx 83.$$

Trap: a mean of Weibull observations is not itself generally Weibull. This calculation is approximate; a "30 observations" rule does not guarantee accurate tail probabilities. Normal populations give exact normal sample means; here the CLT is doing the work.

28. Shifted Weibull and age-replacement cost

Recognise: replace at failure or scheduled age, whichever happens first. **Source:** Week 5 Quiz Q5.

A cable has shape $\beta = 1.8$, scale $\eta = 500$ h and threshold $\gamma = 50$ h. Corrective replacement costs \$6000, scheduled replacement \$600; operation is $H = 5000$ h/year. Candidate ages are $50, 75, \ldots, 300$ h.

$$R(t) = \begin{cases} 1 & t < 50 \\ e^{-\left(\frac{t-50}{500}\right)^{\{1.8\}}} & t \geq 50. \end{cases}$$

Each replacement restores the cable to new condition. Ignoring downtime, the long-run annual cost is

$$C(m) = 5000\frac{6000[1 - R(m)] + 600R(m)}{\int_0^m R(t)\,\mathrm{dt}}.$$

At $m = 175$ h, $F(m) = 0.07916024$ and expected cycle length is 171.40895 h. Therefore

$$C(175) = 5000\frac{6000(0.07916024) + 600(0.92083976)}{171.40895} \approx \$29{,}971.17 \text{ per year.}$$

Candidate age (h)	Annual cost, rounded to nearest dollar
50	60,000
75	41,657
100	34,342
125	31,205
150	30,053
175	**29,971**
200	30,510
225	31,425
250	32,571
275	33,860
300	35,231

Thus 175 h is the minimum **among the specified candidates**. Compute the whole grid, not only neighbouring values. For accurate numerical integration, split at the threshold 50 or use tight tolerances.

Trap: dividing by scheduled age m ignores cycles ending early at failure. A fixed calendar/block replacement policy has a different cost model. Do not claim this grid optimum is the unconstrained continuous optimum.

29. Engineering calculations and short-answer patterns

Recognise: the task asks for an engineering decision, financial metric or classification. **Sources:** 2025 Q1, Q5–7, Q9, Q12 see engineering notes for full slide coverage and further case examples.

Discounted maintenance costs: four end-of-year payments of \$150000 at an 8% real discount rate give

$$\mathrm{NPC} = 150000\sum_{t=1}^{4}(1.08)^{-t} = 496819.03, \qquad \$496{,}819.$$

A time-zero investment would be added without discounting. Positive costs are summed for NPC; NPV requires consistent signs for cash inflows and outflows.

Toy availability calculation: an asset averages 100 operating hours between failures and 5 repair hours per failure. Its long-run availability is

$$A = \frac{100}{100 + 5} = 0.952381.$$

Availability allows repair and return to operation; mission reliability is the probability of no failure over a specified duration.

Short-answer examples:

Problem type	Worked reasoning → answer	Trap
Classify a control	Sealing nitrocellulose packaging physically retains its damping agent → **engineering control**	A written instruction to check the seal would be administrative
Failure vocabulary	"Engine fails to start" describes lost function → **failure mode**; a flat battery can be a cause	Mode, cause and mechanism answer different questions
Prioritise FMEA	Cable connections have $S = 6, O = 7, D = 4$: RPN $= 168$, exceeding leakage's $7(6)(3) = 126$ → select cable connections in this question	RPN is an ordinal prioritisation tool, not failure probability or expected monetary loss
Select maintenance	Vibration/infrared evidence of deterioration → **condition-based maintenance**	The measuring technique supports the tactic; it does not itself define a fixed replacement age
Rate a scenario	In the supplied paper's matrix, event heard of in industry → likelihood B; multiple fatalities → consequence 5; answer **B5**	Use the provided matrix definitions, not invented numerical probabilities
Hidden protection failure	**Toy:** a standby trip can fail undetected until demanded; a feasible proof test reveals loss of protection → **failure-finding task**	A hidden failed device may show no developing P–F warning
Explain a P–F interval	**Toy:** detectable deterioration at day 10, functional failure at day 30 → interval 20 days; inspect sufficiently often to detect and act within it	Inspection spacing alone is insufficient if repair preparation takes longer than the remaining warning time

Engineering answers should connect the observed fact to the definition, then state the decision. Where a decision uses failure probability, hazard, reliability, availability, expected cost or uncertainty, the statistical calculations in this document provide the quantitative input.

30. Point mass and continuous lifetime in one model

Recognise: some items fail immediately; the others have a continuous lifetime. **Toy extension. New step:** retain a CDF jump separately from the continuous density.

A device has probability 0.1 of failing at start-up, recorded as $T = 0$. Conditional on starting successfully, its lifetime is uniform between 1 and 5 hours. Find the CDF, the probability of failure by 2 h, and the mean life.

The start-up failures create a jump of 0.1 at zero. The remaining probability 0.9 is spread over $(1, 5)$:

$$F_{T(t)} = \begin{cases} 0 & t < 0 \\ 0.1 & 0 \le t < 1 \\ 0.1 + 0.9\frac{t-1}{4} & 1 \le t < 5 \\ 1 & t \ge 5. \end{cases}$$

Thus $P(T = 0) = F_{T(0)} - F_{T(0^-)} = 0.1$, and

$$P(T \le 2) = 0.1 + 0.9\frac{2-1}{4} = 0.325.$$

But $P(0 < T \le 2) = 0.325 - 0.1 = 0.225$: including the endpoint zero matters here.

The mean averages the two parts:

$$E[T] = 0.1(0) + 0.9\frac{1+5}{2} = 2.7 \text{ h}.$$

Trap: differentiating the CDF gives density 0.225 on $(1, 5)$, whose area is only 0.9. The missing 0.1 is the atom at zero; an ordinary PDF alone cannot describe this whole distribution.

31. A decreasing nonlinear transformation

Recognise: the required variable is the reciprocal of another random measurement. **Toy extension. New step:** transform the event, reverse the inequality, and determine the new support.

A vehicle travels 12 km at a random constant speed $V \sim U(2, 4)$ km/h. Find the distribution and mean of travel time $T = \frac{12}{V}$.

The fastest and slowest speeds give $3 \le T \le 6$ h. For $3 < t < 6$,

$$F_{T(t)} = P\left(\frac{12}{V} \le t\right) = P\left(V \ge \frac{12}{t}\right) = \frac{4 - \frac{12}{t}}{4-2} = 2 - \frac{6}{t}.$$

Outside the support, $F_{T(t)} = 0$ for $t \le 3$ and $F_{T(t)} = 1$ for $t \ge 6$. Differentiating within it gives

$$f_{T(t)} = \frac{6}{t^2}, \qquad 3 < t < 6.$$

For example, $P(T \le 4) = 2 - \frac{6}{4} = 0.5$. The mean is

$$E[T] = \int_3^6 t\frac{6}{t^2}\,\mathrm{dt} = 6\ln\left(\frac{6}{3}\right) = 4.158883 \text{ h}.$$

Trap: $\frac{12}{E[V]} = \frac{12}{3} = 4$ h is not $E\left[\frac{12}{V}\right]$. Nonlinear transformations generally cannot be applied directly to a mean. A uniform speed also does not imply a uniform travel time.

32. Recover lognormal parameters from original-scale moments

Recognise: a lognormal model is specified using the ordinary mean and SD, rather than `meanlog` and `sdlog`. **Toy extension. New step:** invert the moment equations before calculating a probability.

Repair time T is lognormal with mean 10 h and SD 6 h. Find the parameters of $\ln T$ and the chance that repair exceeds 20 h. Take logarithms of the numerical time measured in hours.

Write $\ln T \sim N(\mu, \sigma^2)$. If $m = E[T]$ and $s = \mathrm{SD}(T)$, then

$$m = e^{\mu + \frac{\sigma^2}{2}}, \qquad \frac{s^2}{m^2} = e^{\sigma^2} - 1.$$

Solve the second equation first, then substitute into the first:

$$\sigma^2 = \ln\left(1 + \frac{6^2}{10^2}\right) = \ln 1.36, \qquad \sigma = 0.554513,$$

$$\mu = \ln 10 - \frac{1}{2}\ln 1.36 = 2.148843.$$

Now the upper tail is

$$P(T > 20) = 1 - \Phi\left(\frac{\ln 20 - 2.148843}{0.554513}\right) = 0.063347.$$

```
sdlog <- sqrt(log(1 + (6 / 10)^2))
meanlog <- log(10) - sdlog^2 / 2
plnorm(20, meanlog = meanlog, sdlog = sdlog, lower.tail = FALSE)
```

Trap: passing `meanlog = 10, sdlog = 6` describes a completely different repair-time model. Changing time units shifts `meanlog`; it does not change `sdlog`.

33. Total variance for a mixture of repair types

Recognise: observations come from categories with different means and spreads. **Toy extension. New step:** include both variation within categories and variation between category means.

Seventy percent of repairs are minor, with mean 2 h and SD 1 h. The other 30% are major, with mean 10 h and SD 3 h. Find the overall mean and SD. No distribution shape within either category is assumed.

Let G denote the repair category. First,

$$E[T] = E[E(T \mid G)] = 0.7(2) + 0.3(10) = 4.4 \text{ h}.$$

The law of total variance is

$$\mathrm{Var}(T) = E[\mathrm{Var}(T \mid G)] + \mathrm{Var}(E[T \mid G]).$$

The within-category contribution is $0.7(1^2) + 0.3(3^2) = 3.4$ h^2. The between-category contribution is

$$0.7(2 - 4.4)^2 + 0.3(10 - 4.4)^2 = 13.44 \text{ h}^2.$$

Therefore,

$$\mathrm{Var}(T) = 16.84 \text{ h}^2, \qquad \mathrm{SD}(T) = \sqrt{16.84} = 4.103657 \text{ h}.$$

Trap: averaging the two variances gives only 3.4 and misses most of the spread. Averaging the two SDs is also incorrect. This is a randomly selected repair from a mixture, not a weighted sum of two repairs.

34. Thinning a Poisson stream

Recognise: a random number of events occurs, and each event independently receives a label. **Toy extension. New step:** combine a Poisson event count with random classification to derive a new count distribution.

Alarms arrive as a Poisson process at 4 per hour. Each alarm is independently serious with probability $p = 0.1$, independently of arrival times. Let K count serious alarms in two hours. Find its distribution and $P(K \geq 1)$.

The total alarm count is $N \sim \text{Poisson}(8)$. Conditional on $N = n$, $K \mid N = n \sim \text{Binomial}(n, p)$. Remove the condition by summing over all possible totals:

$$\begin{aligned} P(K = k) &= \sum_{n=k}^{\infty} \binom{n}{k} p^{k(1-p)^{n-k}} \frac{8^n e^{-8}}{n!} \\ &= \frac{(8p)^k e^{-8}}{k!} \sum_{j=0}^{\infty} \frac{[8(1-p)]^j}{j!} \\ &= \frac{(8p)^k e^{-8p}}{k!}. \end{aligned}$$

Thus $K \sim \text{Poisson}(0.8)$, and

$$P(K \geq 1) = 1 - e^{-0.8} = 0.550671.$$

Trap: $\text{Binomial}(8, 0.1)$ would treat the expected total of eight alarms as a fixed total. The classification must be independent with a constant probability for this thinning result to apply.

35. Competing failure modes and cause-specific probability

Recognise: more than one mode can end a life, and the question asks which one acts first. **Toy extension. New step:** multiply one mode's failure density by survival against the competing mode, then integrate.

An item can fail electrically or mechanically. Its independent latent failure times are exponential with rates $\lambda_E = 0.002\ \text{h}^{-1}$ and $\lambda_M = 0.003\ \text{h}^{-1}$. The first failure ends service. Find the probability that electrical failure ends service by 100 h.

For electrical failure at age u to be the observed cause, the mechanical mode must not yet have occurred. Hence

$$\begin{aligned} P(T_E < T_M,\ & T_E \leq 100) = \int_0^{100} f_{E(u)} R_{M(u)}\, \text{du} \\ &= \int_0^{100} 0.002 e^{-0.002u} e^{-0.003u}\, \text{du} \\ &= \frac{0.002}{0.005}\left(1 - e^{-0.005(100)}\right) \\ &= 0.157388. \end{aligned}$$

Over an unlimited time horizon, electrical failure wins the race with probability $\frac{\lambda_E}{\lambda_E + \lambda_M} = 0.4$. In this constant-rate model, the same fraction 0.4 of failures observed by 100 h are electrical, since dividing by $P(\min(T_E, T_M) \leq 100)$ cancels the bracket.

Trap: $1 - e^{-0.002(100)}$ ignores items already removed by mechanical failure. Cause-specific cumulative failure probability depends on all competing hazards; the constant rate-ratio answer is not a general rule for time-varying hazards.

36. Common-cause failure and the limit of redundancy

Recognise: a shared event can disable all redundant units. **Toy extension. New step:** model dependence explicitly and identify the reliability ceiling.

Two backup pumps serve one mission. A shared flood disables both with probability $q = 0.02$. Conditional on no flood, their individual failure probabilities are each $p = 0.1$, independently. Find the probability both are unavailable and the best reliability achievable by adding more such pumps.

Both pumps are unavailable either because of the flood or because both fail individually without a flood. These cases are disjoint:

$$P(\text{system fails}) = q + (1-q)p^2 = 0.02 + 0.98(0.1)^2 = 0.0298.$$

Thus system reliability is 0.9702. Each pump's marginal failure probability is $q + (1-q)p = 0.118$, but multiplying these marginals gives $0.118^2 = 0.013924$, which is too low because the pumps are dependent.

For n pumps,

$$R_n = (1-q)(1-p^n), \qquad \lim_{n\to\infty} R_n = 1 - q = 0.98.$$

A target of 0.99 is therefore impossible by adding pumps alone under this model; the shared flood vulnerability must also change.

Trap: physical duplication does not establish statistical independence. The stated individual probability p is conditional on no flood, so the flood contribution must be accounted for separately.

37. Two-out-of-three with unequal components

Recognise: at least two units must work, but they have different success probabilities. **Toy extension. New step:** enumerate disjoint functioning states instead of substituting into an identical-trial binomial model.

Three independent sensors have mission reliabilities $R_A = 0.9$, $R_B = 0.8$ and $R_C = 0.7$. The system requires any two functioning sensors. Find system reliability.

There are four successful states:

Working sensors	State probability
A, B only	$0.9(0.8)(0.3) = 0.216$
A, C only	$0.9(0.7)(0.2) = 0.126$
B, C only	$0.8(0.7)(0.1) = 0.056$
All three	$0.9(0.8)(0.7) = 0.504$

Add the disjoint states:

$$R_{\text{sys}} = 0.216 + 0.126 + 0.056 + 0.504 = 0.902.$$

Equivalently,

$$R_{\text{sys}} = R_A R_B + R_A R_C + R_B R_C - 2R_A R_B R_C.$$

The subtraction removes two extra counts of the all-working state. Using the average reliability 0.8 in a binomial formula would give $3(0.8)^2(0.2) + (0.8)^3 = 0.896$, not the exact answer.

Trap: "at least two work" is neither ordinary series nor ordinary parallel. State explicitly what counts as system success before calculating.

38. Cold standby and imperfect switching

Recognise: one unit operates while a spare waits without ageing or failing. **Toy extension. New step:** integrate over the time of transfer to the spare.

An operating unit and a cold spare each have independent exponential operating lives with rate $\lambda = 0.001\ \text{h}^{-1}$. The spare cannot fail while waiting. There is no repair during the mission. Find reliability at 1000 h, first with perfect switching and then with switching success probability $c = 0.95$ independent of the lifetimes.

The system survives to t if the first unit lasts throughout, or if it fails at some $u < t$, transfer succeeds, and the spare lasts the remaining $t - u$:

$$\begin{aligned} R_{\text{standby}}(t) &= e^{-\lambda t} + c\int_0^t \lambda e^{-\lambda u} e^{-\lambda(t-u)}\,\mathrm{du} \\ &= e^{-\lambda t}(1 + c\lambda t). \end{aligned}$$

At $t = 1000$ h,

$$c = 1: \quad R = 2e^{-1} = 0.735759, \qquad c = 0.95: \quad R = 1.95e^{-1} = 0.717365.$$

With perfect switching, total life is the **sum** of two operating lives. By comparison, if both units age and operate from time zero, active-parallel reliability is $2e^{-1} - e^{-2} = 0.600424$.

Trap: cold standby does not follow the active-parallel formula. Spare deterioration, switch failure, transfer delay or load changes require their own model assumptions.

39. Design the number of redundant units

Recognise: the number of components is the unknown, with an integer reliability requirement. **Toy extension. New step:** invert the parallel formula, handle the logarithm's sign, and check the neighbouring integer.

Independent identical units each have mission reliability $r = 0.85$. At least one must survive. Find the smallest total number n that achieves system reliability at least 0.999.

$$1 - (1 - r)^n \geq 0.999 \quad \Leftrightarrow \quad 0.15^n \leq 0.001.$$

Taking logarithms gives $n \ln 0.15 \leq \ln 0.001$. Since $\ln 0.15 < 0$, division reverses the inequality:

$$n \geq \frac{\ln 0.001}{\ln 0.15} = 3.641180.$$

Thus $n_{\min} = 4$. Verify minimality:

$$R_3 = 1 - 0.15^3 = 0.996625 < 0.999, \qquad R_4 = 1 - 0.15^4 = 0.99949375 \geq 0.999.$$

Trap: n counts all units, including the original one. This design needs three added spares if one unit already exists. Shared failures invalidate the independence calculation; check the ceiling in Example 36 before solving for a larger n.

40. Survivor selection in a mixed fleet

Recognise: the fleet contains persistent subgroups with different hazards. **Toy extension. New step:** weight subgroup hazards by the composition of the surviving population, which changes with age.

Forty percent of new units are a fragile type with exponential rate $0.02\ \text{h}^{-1}$; 60% are a robust type with rate $0.002\ \text{h}^{-1}$. A unit stays in its original type. Find the fleet hazard at ages 0 and 100 h.

The fleet survival and density are

$$R(t) = 0.4e^{-0.02t} + 0.6e^{-0.002t},$$

$$f(t) = 0.4(0.02)e^{-0.02t} + 0.6(0.002)e^{-0.002t}.$$

Initially $h(0) = 0.4(0.02) + 0.6(0.002) = 0.0092\ \text{h}^{-1}$. At 100 h, only

$$w_{F(100)} = \frac{0.4e^{-2}}{0.4e^{-2} + 0.6e^{-0.2}} = 0.099261$$

of survivors are fragile. Therefore,

$$h(100) = w_{F(100)}(0.02) + \left[1 - w_{F(100)}\right](0.002) = 0.00378669\ \text{h}^{-1}.$$

The fleet hazard falls because fragile units disappear faster. Neither subgroup's individual hazard improves with age. As $t \to \infty$, the fleet hazard approaches the robust rate $0.002\ \text{h}^{-1}$.

Trap: continuing to use the original 40%/60% weights at every age is wrong. A mixture of exponentials need not itself be exponential, and a falling aggregate hazard does not by itself prove that individual items become more reliable as they age.

41. Mean remaining life

Recognise: the question asks for expected additional life after an item has survived, rather than a survival probability at one future time. **Toy extension. New step:** integrate the entire conditional remaining-life tail.

A lifetime has survival $R(t) = \left(1 + \frac{t}{100}\right)^{-3}$ for $t \geq 0$, with t in hours. An item is still working at age 200 h. Find its expected remaining life.

For age a, define $Y = T - a$ conditional on $T > a$. Its tail is $P(Y > u \mid T > a) = \frac{R(a+u)}{R(a)}$. Consequently,

$$m(a) = E[T - a \mid T > a] = \int_0^\infty \frac{R(a+u)}{R(a)}\,\text{du} = \frac{\int_a^\infty R(t)\,\text{dt}}{R(a)}.$$

Here,

$$\int_a^{\infty\left(1+\frac{t}{100}\right)^{-3}} \text{dt} = 50\left(1 + \frac{a}{100}\right)^{-2},$$

so

$$m(a) = 50\left(1 + \frac{a}{100}\right) = \frac{100 + a}{2}.$$

At $a = 200$, $m(200) = 150$ h, and expected total life given survival is $200 + 150 = 350$ h.

Trap: the unconditional mean is only $m(0) = 50$ h, but $50 - 200 = -150$ h is not a remaining-life estimate. Surviving to 200 h supplies information; conditioning must come before taking the expectation.

42. Interval censoring and the likelihood

Recognise: an inspection finds an item failed, but its exact failure time is unknown. **Toy extension. New step:** use probability over an interval as the likelihood contribution.

Ten independent new components have exponential lifetimes with rate λ. They are inspected once at 10 h. Two have failed sometime in $(0, 10]$; eight remain working. Estimate λ from exactly this information.

Each failed unit contributes

$$P(0 < T \le 10) = F(10) - F(0) = 1 - e^{-10\lambda},$$

and each survivor contributes $R(10) = e^{-10\lambda}$. Ignoring a factor constant in λ,

$$L(\lambda) \propto \left(1 - e^{-10\lambda}\right)^2 \left(e^{-10\lambda}\right)^8.$$

Set $s = e^{-10\lambda}$, the survival probability to inspection. Then

$$\frac{d}{\mathrm{d}s} \ln L = -\frac{2}{1-s} + \frac{8}{s} = 0 \quad \Rightarrow \quad \hat{s} = 0.8.$$

Thus

$$\hat{\lambda} = -\frac{\ln 0.8}{10} = 0.0223144 \text{ h}^{-1}.$$

More generally, failure known only to lie in $(L, U]$ contributes $F(U) - F(L) = R(L) - R(U)$. An exact continuous failure time contributes $f(t)$; right-censoring contributes $R(t)$.

Trap: recording both failures as occurring exactly at 10 h would give the wrong likelihood and rate $\frac{2}{100} = 0.02$. Substituting interval midpoints also discards the actual observation mechanism.

43. Delayed entry changes the risk set

Recognise: units first become observable after different amounts of life have already elapsed. **Toy extension beyond the basic KM exercises. New step:** exclude units from risk sets before their entry ages.

The following records use **age since installation**, not time since joining the study. Assume entry and censoring are noninformative for lifetime estimation.

Unit	Entry age (h)	Exit age (h)	Exit outcome
A	0	4	Failure
B	0	7	Censored
C	2	5	Failure
D	4.5	8	Failure
E	6	9	Censored

At failure age t, a unit is at risk if it has entered and has not already exited. There are no tied entry/failure ages in this dataset.

Failure age	Units at risk	Survival update
4	A, B, C	$1\left(1-\frac{1}{3}\right)=\frac{2}{3}$
5	B, C, D	$\left(\frac{2}{3}\right)\left(1-\frac{1}{3}\right)=\frac{4}{9}$
8	D, E	$\left(\frac{4}{9}\right)\left(1-\frac{1}{2}\right)=\frac{2}{9}$

Hence $\hat{R}(6)=\frac{4}{9}$ and $\hat{R}(8)=\frac{2}{9}$. Entry itself does not cause a jump, and neither does B's censoring at age 7.

```r
library(survival)
entry <- c(0, 0, 2, 4.5, 6)
exit_age <- c(4, 7, 5, 8, 9)
failed <- c(1, 0, 1, 1, 0)
fit_entry <- survfit(Surv(entry, exit_age, failed) ~ 1)
summary(fit_entry, times = c(6, 8))$surv
# 0.4444444 0.2222222
```

Trap: counting all five units at age 4 gives credit for survival to units not yet observed. Resetting every entry age to zero also changes the time scale. Delayed entry needs its own risk-set accounting in addition to ordinary right-censoring.

44. Restricted mean from a survival curve

Recognise: follow-up ends while estimated survival is still positive. **Toy extension. New step:** estimate expected life up to a specified horizon by adding rectangular areas under the survival curve.

Suppose an estimated survival curve has the following values through the last observation at 10 h:

Age interval (h)	$\widehat{R}(t)$
$[0,2)$	1
$[2,5)$	0.8
$[5,8)$	0.5
$[8,10]$	0.25

Find the restricted mean lifetime to $\tau=10$ h. This estimates $E[\min(T,10)]$, where an item lasting beyond 10 contributes 10 hours:

$$\begin{aligned}\widehat{\mathrm{RMST}}(10) &= \int_0^{10} \hat{R}(t)\,\mathrm{dt} \\ &= 2(1)+3(0.8)+3(0.5)+2(0.25) \\ &= 6.4 \text{ h}.\end{aligned}$$

It is not $E[T \mid T \le 10]$: the restricted mean includes the first 10 h of those who outlive the horizon. It is also not the full MTTF, whose integral continues beyond 10 h.

Trap: setting survival to zero after follow-up invents an unobserved terminal failure. Keeping a positive final KM step forever gives an infinite integral without establishing an infinite population mean. The area to the supported horizon is identifiable without inventing the tail.

45. Zero failures and a test-exposure requirement

Recognise: all tested items survive, but a finite reliability statement is still required. **Toy extension beyond point estimation. New step:** invert the probability of a zero-failure result to obtain a one-sided confidence bound.

Twenty independent components each complete a fixed 1000-hour test with zero failures. Assume identical exponential lifetimes. Find a one-sided 95% upper confidence bound for the failure rate, then design the exposure needed to make that bound at most 10^{-4} h^{-1} if no failures occur.

The observed total exposure is $E = 20000$ component-hours. Under rate λ, the probability that all survive their fixed tests is

$$P(\text{zero failures}|\ \lambda) = e^{-\lambda E}.$$

For this zero-failure outcome, the one-sided upper bound solves $e^{-\lambda_U E} = \alpha$ with $\alpha = 0.05$:

$$\lambda_U = -\frac{\ln 0.05}{20000} = 1.497866 \times 10^{-4}\ \mathrm{h}^{-1}.$$

The corresponding 95% lower confidence bound on mean life is $\frac{1}{\lambda_U} = 6676.16$ h. The MLE $\hat{\lambda} = 0$ alone would provide no finite upper estimate of mean life.

For the design target,

$$\frac{-\ln 0.05}{E} \le 10^{-4} \quad \Rightarrow \quad E \ge 29957.32 \text{ component-hours}.$$

With fixed 1000-hour tests, use at least $\lceil 29.95732 \rceil = 30$ independent units; the claimed bound requires zero observed failures.

Trap: zero observed failures does not establish a true zero failure rate. A 95% confidence bound describes a repeated-sampling procedure, not a posterior 95% probability for the parameter. Pooling component-hours this way depends on the common constant-hazard model and the fixed test design.

46. A distribution-free bound for a sample mean

Recognise: a rigorous bound is requested, but the population distribution is unspecified. **Toy extension beyond the CLT exercises. New step:** use an inequality rather than an approximate sampling distribution.

Independent identically distributed measurements have finite mean μ and known SD $\sigma = 12$. Find a sufficient sample size to guarantee

$$P\big(|\bar{X} - \mu| < 4\big) \ge 0.99$$

without assuming normality or relying on a CLT approximation.

Chebyshev's inequality states $P(|Y - E[Y]| \ge d) \le \mathrm{Var}\,\frac{Y}{d^2}$. Since $\mathrm{Var}\big(\bar{X}\big) = \frac{144}{n}$,

$$P\big(|\bar{X} - \mu| \ge 4\big) \le \frac{\frac{144}{n}}{16} = \frac{9}{n}.$$

It is sufficient to make $\frac{9}{n} \le 0.01$, giving

$$n \ge 900.$$

For comparison, a normal-population or CLT design using the same moments gives $n = 60$. The much larger bound pays for protection across arbitrary population shapes with those moments.

Trap: 900 is sufficient under Chebyshev's bound; this calculation does not prove it is the smallest possible universal sample size. A CLT approximation and a distribution-free guarantee answer different requirements. Replacing known σ by an uncertain sample SD does not preserve this guarantee automatically.

47. Stress-strength interference

Recognise: both the load and the strength are random, and failure occurs when load exceeds strength. **Toy extension. New step:** integrate conditional failure over the overlapping supports of two distributions.

Let independent strength $S \sim U(80, 120)$ kN and load $L \sim U(60, 100)$ kN. Find $P(L > S)$.

For strength $s \geq 100$, the load cannot exceed it. For $80 < s < 100$,

$$P(L > s \mid S = s) = P(L > s) = \frac{100 - s}{40}.$$

Average this over the strength density $f_{S(s)} = \frac{1}{40}$:

$$\begin{aligned} P(L > S) &= \int_{80}^{100} \frac{100 - s}{40} \frac{1}{40} \, \mathrm{ds} \\ &= \frac{1}{1600} \left[100s - \frac{s^2}{2} \right]_{80}^{100} \\ &= 0.125. \end{aligned}$$

Equivalently, the failure region is a triangle of area $\frac{1}{2}(20)(20) = 200$ inside a joint-support rectangle of area $40(40) = 1600$. Mission success probability is $P(S > L) = 0.875$; equality has probability zero.

Trap: comparing means, $E[S] = 100$ and $E[L] = 80$, does not calculate failure probability. Independence is what permits using the marginal load tail inside the integral.

48. Proof-test spacing and average unavailability

Recognise: a protective device can fail silently and remains failed until its next test. **Toy extension. New step:** average hidden-failure probability over all times within the testing cycle.

A trip device is restored to working condition at every perfect proof test, spaced τ hours apart. Between tests its constant hidden-failure rate is $\lambda = 0.0002\ \mathrm{h}^{-1}$. Tests and restoration take negligible time. Demands are independent and equally likely at any point in the interval. Find average unavailability for $\tau = 100$ h and the largest interval giving at most 0.005 average unavailability.

At age u since the last test, the device is unavailable with probability $1 - e^{-\lambda u}$. For a uniformly timed demand,

$$\overline{Q}(\tau) = \frac{1}{\tau} \int_0^{\tau(1 - e^{-\lambda u})} \mathrm{du} = 1 - \frac{1 - e^{-\lambda \tau}}{\lambda \tau}.$$

For $\tau = 100$ h,

$$\overline{Q}(100) = 1 - \frac{1 - e^{-0.02}}{0.02} = 0.00993367.$$

For small $\lambda\tau$, expand the exponential to get $\overline{Q}(\tau) \approx \lambda \frac{\tau}{2}$. This gives an initial design estimate $\tau \approx 2\frac{0.005}{0.0002} = 50$ h. Solve the exact expression for the boundary:

```
average_Q <- function(tau) {
  1 - (-expm1(-0.0002 * tau)) / (0.0002 * tau)
}
uniroot(function(tau) average_Q(tau) - 0.005, c(1, 100))$root
# 50.16736 hours
```

Since average unavailability increases with interval length, the requirement is $\tau \leq 50.16736$ h; 50 h meets it.

Trap: end-of-interval unavailability $1 - e^{-\lambda\tau}$ is not the average seen by randomly timed demands. The factor-of-two approximation depends on small $\lambda\tau$ and the stated perfect-test assumptions.

49. Inspection timing with a repair lead time

Recognise: deterioration must be detected early enough to finish an intervention. **Toy extension. New step:** treat the delay to the next inspection as a random variable and reserve time for the repair.

Once deterioration becomes detectable, functional failure follows exactly $D = 20$ days later. Inspections occur every $\Delta = 14$ days and always detect existing deterioration. Detection-to-completed-repair lead time is $L = 8$ days. Assume the onset of detectable deterioration is uniformly located within an inspection interval. Find the probability of preventing failure and the largest interval meeting a 95% prevention target.

Under the random-onset assumption, the delay W from detectable deterioration to the next inspection is uniform on $[0, \Delta]$. Successful intervention requires

$$W + L < D \quad \Leftrightarrow \quad W < D - L.$$

Thus

$$P(\text{prevent failure}) = \begin{cases} 0 & D \leq L \\ \min\left\{1, \frac{D-L}{\Delta}\right\} & D > L. \end{cases}$$

For the proposed interval,

$$P(\text{prevent failure}) = \frac{20 - 8}{14} = 0.857143.$$

To reach 95%, require $\frac{12}{\Delta} \geq 0.95$, so $\Delta \leq 12.63158$ days. A 12-day interval gives probability one under the continuous-phase model; a strict worst-case margin requires $\Delta + L < D$, so $\Delta < 12$ days.

Trap: choosing an inspection interval merely shorter than the 20-day P–F interval ignores the eight-day repair lead time. Imperfect detection or uncertain deterioration/repair times would change these answers.

50. F–N exceedance frequencies and potential loss of life

Recognise: scenarios have annual frequencies and different fatality counts. **Toy extension connected to the engineering notes. New step:** accumulate frequencies by consequence threshold, then distinguish that curve from expected annual loss.

For a hypothetical facility, three disjoint accident outcome categories have the following frequencies. Treat these as supplied model inputs, not measured risk estimates.

Outcome category	Fatalities per event N_i	Frequency f_i (year^{-1})
A	1	2×10^{-3}
B	5	3×10^{-4}
C	20	1×10^{-5}

Here $F_{N(n)}$ means the annual **frequency of events with at least n fatalities**. Construct it by summing the categories meeting the threshold:

$$F_{N(n)} = \sum_{i : N_i \geq n} f_i .$$

Fatality threshold n	**Categories included**	**$F_{N(n)}$ (year^{-1})**
1	A, B, C	0.00231
5	B, C	0.00031
20	C	0.00001
21	None	0

Potential loss of life (PLL) weights each event frequency by its fatality count:

$$\mathrm{PLL} = \sum_i f_i N_i = 0.002(1) + 0.0003(5) + 0.00001(20) = 0.0037 \text{ expected fatalities/year.}$$

The discrete tail-sum check recovers the same value: $F_{N(1)} + 4F_{N(5)} + 15F_{N(20)} = 0.00231 + 4(0.00031) + 15(0.00001) = 0.0037$. The multiples account for thresholds 2–5 and 6–20, over which the exceedance frequency is constant.

Trap: an F–N curve uses cumulative exceedance frequencies, not each category's isolated frequency. PLL is neither an individual's annual fatality probability nor a guarantee about what happens in a particular year. Scenarios must avoid counting the same accident outcome twice.